Classical plane geometry by coordinate certificates: Desargues, Monge, Simson, Napoleon and Lami in Lean 4

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Abstract

We formalise five classical theorems of plane geometry in Lean 4 with Mathlib: Desargues's theorem in the projective plane, together with its converse; Monge's theorem that the three external centres of similitude of three circles are collinear; the Wallace–Simson theorem on the pedal triangle of a point of the circumcircle; Napoleon's theorem for the outer and the inner Napoleon triangles; and Lami's theorem for three forces in equilibrium. Every proof has the same core. The theorem is reduced to an explicit polynomial or rational identity in coordinates, which we call a *certificate*, and the identity is closed by `ring`, `field_simp` or `linear_combination` with a stated multiplier. Around this core, each development makes precise which non-degeneracy conditions are needed, which are not, and why. Desargues's theorem and Monge's theorem are proved over an arbitrary field, and the Desargues identity over an arbitrary commutative ring. The Simson line and Napoleon's theorem are proved in the Euclidean plane, modelled as $\mathbb{C}$. Lami's theorem is proved through the oriented area form of a two-dimensional oriented real inner product space, and in unoriented form in every real inner product space. The five libraries, without the version-2 files described next, have about 1,430 lines of Lean, use no `sorry`, and depend only on Lean's three standard axioms. Version 2 adds to each library a file giving Mathlib-only formulations of the main theorems (for Napoleon's theorem, of its non-degenerate configurations), each with a non-degenerate witness, and a file proving from the libraries textbook statements written in Lean by an AI model that saw neither our code nor this paper; these include the geometric converse of the Wallace–Simson theorem, which the first version did not prove. The mathematics is classical and the method is old, and several of the results are not new to formal mathematics (Section 9). Monge's three-circle theorem has two earlier, independent and complete Lean proofs, one of which predates ours by four months; Napoleon's theorem has earlier formalisations in Lean, Isabelle and Rocq, and a later independent one in Lean; the Wallace–Simson theorem and its converse are formalised in Isabelle and in Rocq; Desargues's theorem is formalised in at least five other systems, including a coordinate (bracket) proof in HOL Light; and an open, unmerged Mathlib pull request proves the affine (parallel) form of Desargues's theorem. We found no earlier Lean proof of the projective-plane form of Desargues's theorem, of the Wallace–Simson theorem, or of Lami's theorem; these searches are not exhaustive.

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1 Introduction

Elementary plane geometry is a natural testing ground for proof assistants: the statements are short and familiar, but a faithful formal statement has to say exactly which configurations are allowed. Textbooks rarely do. "Two triangles in perspective from a point" may or may not allow coincident vertices; "three circles" may or may not allow equal radii; "the ratio $|F_1|/\sin\alpha$" is not defined when $\sin\alpha = 0$. A formal statement must decide each of these questions, and the decision is part of the result.

The proofs in this paper are coordinate proofs in the tradition of mechanical geometry theorem proving [6]. Each theorem is reduced to an identity of the form

$$(\text{conclusion}) \cdot (\text{non-degenerate factor}) \;=\; \sum_i c_i \cdot (\text{hypothesis}_i),$$

with the multipliers c_i written out (or with no hypotheses at all on the right), and the identity is checked by a decision procedure for commutative rings. We call such an identity a *certificate*. The certificate is only the algebraic core. Each development also has a translation layer connecting the coordinate statement with a standard notion of Mathlib, such as `Projectivization.IsCollinear`, `Collinear`, `dist` or `Orientation.oangle`. That layer is where most of the non-degeneracy hypotheses enter. The point of the paper is the analysis of those hypotheses as much as the identities.

Contributions.

- Desargues's theorem in Mathlib's projective plane $\mathbb{P}(K^3)$ over an arbitrary field K, in a determinant form that needs only a nonzero centre of perspective, and in a projective form whose distinctness hypotheses are exactly those that make Mathlib's `cross` compute sides and intersection points (Section 3). The converse is proved in both forms. The underlying factorisation $[PQR] = [abc]\,[a'b'c']\,[AA'\,BB'\,CC']$ is proved over every commutative ring.
- Monge's theorem in coordinates over an arbitrary field, with pairwise distinct radii as the only hypothesis, and no hypothesis on the centres (Section 4). This theorem is already proved in Lean, twice and independently; our version differs only in being stated over an arbitrary field and in its explicit analysis of the hypotheses.
- The forward direction of the Wallace–Simson theorem in $\mathbb{C}$, with conclusion Mathlib's `Collinear` and hypotheses "the vertices are pairwise distinct" and Mathlib's `EuclideanGeometry.Cospherical`, together with a determinant-level equivalence for a non-degenerate triangle. The first version did not prove the geometric converse; version 2 proves it, in `EuclideanSpace` $\mathbb{R}$ `(Fin 2)`, in a Layer-15 file (`Simson.Classical.cov_A3_b`; Sections 5 and 10.1). This theorem, with its converse, is already formalised in Isabelle and in Rocq.
- Napoleon's theorem in $\mathbb{C}$ as an equality of distances, for both Napoleon triangles and with no non-degeneracy hypothesis, together with a signed-area lemma identifying which triangles are erected outward (Section 6).
- Lami's theorem in oriented and unoriented forms, product and ratio forms, an explicit treatment of the degenerate (parallel) case, and the oriented converse (Section 7).
- A table of the non-degeneracy conditions of all five theorems (Table 2), recording for each whether the certificate needs it, the translation needs it, or it is only needed for the statement to mean what it says.

- (Version 2.) Mathlib-only formulations of the six numbered theorems (for Napoleon, of the non-degenerate configurations), each with a non-degenerate witness, and proofs from the libraries of textbook statements of all five theorems written in Lean by a model that saw neither code nor paper (Section 10.1).

Availability. The five developments are publicly available at the following addresses: `github.com/jpbald93/desargues`, `github.com/jpbald93/monge`, `github.com/jpbald93/simson`, `github.com/jpbald93/napoleon` and `github.com/jpbald93/lami`. Line references (Table 1) are to commits `b640925`, `a01f750`, `7c239b3`, `e16592a` and `ea6b038` respectively, and link to those commits. In the first four repositories that commit adds a module `Bridge.lean` of definition bridges (Section 8) on top of the original content commit and a citation-only commit; apart from one added `import` line at the end of each root file, the earlier files are unchanged. The Lami commit changes only the citation in `README.md`, so its Lean sources are those of its content commit `a7a039b`. Each repository has since received three further commits: one correcting the citation in `README.md`, one making the verification scripts `gate.sh` and `tests/tamper.sh` portable to BSD-derived systems, which had required GNU `mktemp`, `sed` and `python3`, and one adding licence and citation metadata (`LICENSE`, `CITATION.cff`, `README.md`); none changes any Lean source, so the commits cited above remain the sources of the formalisations. Version 2 of this paper adds the Layer-14 and Layer-15 files (Section 10.1) in one new commit of each repository (desargues `086ca718192df1cb919b72a7e79d9fb26e43e723`, monge `82dd27d7b81e60e2aa57873e98cc6fa0dce0728f`, simson `e3d2c46abd6a6da2a4571a28cebc47c8447a0618`, napoleon `0829a1483687956644b6b5eb3eb26205c6d7e9b7`, lami `e692d1a9bc0004444e33b76132275461a2e26cf2`), published on GitHub. That commit adds `L14.lean` and `L15Classical.lean` to the library directory (for example `Simson/L15Classical.lean`), appends their imports to the root file, adds the new declarations to the list checked by `gate.sh`, updates `README.md`, and removes the `scratch/` directory, which is not part of the library. The other library files are unchanged, so the statements, files and line references of the commits cited above are unchanged. The code in all five repositories (Lean sources, gate and tamper scripts, and the Python certificate scripts) is released under the Apache License 2.0, the licence used by Mathlib; this paper is released under CC BY 4.0. The toolchain is Lean `v4.33.1` [7] with Mathlib [14] revision `0df444a3` (tag `v4.33.1`), pinned in each `lake-manifest.json`. The libraries and their sizes are `Desargues` (five files, 452 lines), `Monge` (two files, 173 lines), `Simson` (four files, 320 lines), `Napoleon` (two files, 205 lines) and `Lami` (one file, 284 lines), each with a root file that only imports. Version 2 adds 218 and 258 lines (Desargues), 91 and 261 (Monge), 121 and 168 (Simson), 334 and 503 (Napoleon), and 73 and 98 (Lami) in `L14.lean` and `L15Classical.lean`. All declarations are in the namespace of the library name (the Layer-15 ones in its sub-namespace, e.g. `Simson.Classical`). Table 1 lists the main formal statements. Supporting material (the Layer-14 and Layer-15 reports, the blind writer's statements and report, the transcript audit, and the reviews of this version with the fixes made in response) is in the folder `PaperA-coordinate-certificates` of the repository `https://github.com/jpbald93/LEAN-verification`.

Table 1: Informal statements and their formal counterparts. Line numbers refer to the first line of the declaration. "Field" means an arbitrary field K.

Informal statement	Lean name	Source	Setting / hypotheses
Desargues (Section 3)			

Informal statement	Lean name	Source	Setting / hypotheses
Bracket $[uvw] = u \cdot (v \times w)$	`Desargues.br`	`Basic`:26	comm. ring; definition
$[uvw] = \det(u, v, w)$	`Desargues.br_eq_det`	`Basic`:28	comm. ring; none
Certificate $[PQR] = [abc][a'b'c'][AA'BB'CC']$	`Desargues.desargues_det_identity`	`Basic`:34	comm. ring; none
Bracket $0 \Rightarrow$ collinear in $\mathbb{P}^2$	`Desargues.isCollinear_mk_of_br_eq_zero`	`Plane`:27	field; $u, v, w \neq 0$, $[uvw] = 0$
Collinear in $\mathbb{P}^2 \Rightarrow$ bracket 0	`Desargues.br_eq_zero_of_isCollinear_mk`	`Plane`:59	field; $u, v, w \neq 0$, `IsCollinear`
Common orthogonal $o \neq 0 \Rightarrow$ bracket 0	`Desargues.br_eq_zero_of_dotProduct_eq_zero`	`Plane`:88	field; $o \neq 0$, $o \cdot x = o \cdot y = o \cdot z = 0$
Bracket $0 \Rightarrow$ common orthogonal $o \neq 0$	`Desargues.exists_dotProduct_eq_zero_of_br_eq_zero`	`Plane`:100	field; $[xyz] = 0$
Desargues, determinant form (Thm. 1)	`Desargues.desargues_det`	`Main`:33	field; $o \neq 0$, $[oaa'] = [obb'] = [occ'] = 0$
Desargues in $\mathbb{P}^2$ (Thm. 2)	`Desargues.desargues_mk`	`Main`:59	field, decidable eq.; $a, b, c, a', b', c', o \neq 0$; six vertex inequalities $A \neq B$, $B \neq C$, $C \neq A$, $A' \neq B'$, $B' \neq C'$, $C' \neq A'$; $[oaa'] = [obb'] = [occ'] = 0$; $AB \neq A'B'$, $BC \neq B'C'$, $CA \neq C'A'$
Converse, determinant form	`Desargues.desargues_converse_det`	`Main`:88	field; $[abc] \neq 0$, $[a'b'c'] \neq 0$, $[PQR] = 0$
Converse in $\mathbb{P}^2$ (dual form)	`Desargues.desargues_converse_mk`	`Main`:106	field, decidable eq.; $a \times a', b \times b', c \times c' \neq 0$; $[abc], [a'b'c'] \neq 0$; $P, Q, R \neq 0$; $\{P, Q, R\}$ `IsCollinear`
Non-degeneracy of the example	`Desargues.ex_nonvacuous`	`Examples`:41	$K = \mathbb{Q}$; none
Desargues applied to the example	`Desargues.ex_desargues`	`Examples`:74	$K = \mathbb{Q}$; none
Monge (Section 4)			
External centre of similitude	`Monge.ext`	`Basic`:31	field; definition
Signed double area	`Monge.det3`	`Basic`:36	field; definition
Monge's theorem (Thm. 3)	`Monge.monge`	`Basic`:42	field; $r_1 \neq r_2$, $r_2 \neq r_3$, $r_3 \neq r_1$
Non-vacuity example over $\mathbb{Q}$	`Monge.monge_example`	`Basic`:54	$K = \mathbb{Q}$; none
Values of the example's centres	`Monge.monge_example_values`	`Basic`:61	$K = \mathbb{Q}$; none
Simson (Section 5)			
Foot of perpendicular	`Simson.perpFoot`	`Basic`:28	$\mathbb{C}$; definition
Foot lies on the line	`Simson.perpFoot_mem_line`	`Basic`:50	$\mathbb{C}$; none

Informal statement	Lean name	Source	Setting / hypotheses
Foot is orthogonal	`Simson.inner_perpFoot`	`Basic`:54	$\mathbb{C}$; $u \neq v$
Concyclicity determinant	`Simson.concyc`	`Identity`:16	$\mathbb{C}$; definition
Concyclic $\Rightarrow$ determinant 0	`Simson.concyc_eq_zero_of_dist`	`Identity`:27	$\mathbb{C}$; $\lvert a-o\rvert = \lvert b-o\rvert = \lvert c-o\rvert = \lvert p-o\rvert = r$
Certificate (pedal area identity)	`Simson.area2_perpFoot_mul`	`Identity`:43	$\mathbb{C}$; $a \neq b$, $b \neq c$, $c \neq a$
Simson, area form	`Simson.simson_area`	`Main`:51	$\mathbb{C}$; a, b, c pairwise distinct; $\lvert a-o\rvert = \lvert b-o\rvert = \lvert c-o\rvert = \lvert p-o\rvert = r$
Simson line, forward direction (Thm. 4)	`Simson.simson`	`Main`:64	$\mathbb{C}$; a, b, c pairwise distinct; $\{a, b, c, p\}$ `Cospherical`
Determinant-level equivalence (not the geometric converse)	`Simson.area2_perpFoot_eq_zero_iff`	`Main`:74	$\mathbb{C}$; $\mathrm{area2}(a, b, c) \neq 0$
Napoleon (Section 6)			
Apex of erected triangle	`Napoleon.apex`	`Basic`:28	$\mathbb{C}$; definition
Erected triangle is equilateral	`Napoleon.apex_equilateral`	`Basic`:64	$\lvert\omega\rvert = 1$, $\lvert 1-\omega\rvert = 1$
Certificate $G_b - G_a = \omega(G_c - G_a)$	`Napoleon.key`	`Basic`:74	$\omega^2 - \omega + 1 = 0$
Napoleon, general ω	`Napoleon.napoleon_general`	`Basic`:81	$\omega^2 - \omega + 1 = 0$, $\lvert\omega\rvert = 1$, $\lvert 1-\omega\rvert = 1$ (redundant, see text); no non-degeneracy
Outer Napoleon triangle (Thm. 5)	`Napoleon.napoleon_outer`	`Basic`:98	none
Inner Napoleon triangle	`Napoleon.napoleon_inner`	`Basic`:104	none
Signed area of erected triangle is $-\frac{\sqrt{3}}{4}\lvert Q-P\rvert^2$	`Napoleon.apex_outward`	`Basic`:112	none (clockwise only if $P \neq Q$)
Lami (Section 7); V real inner product space, $\dim V = 2$ *and oriented where stated*			
Equilibrium $\Rightarrow$ equal cyclic area forms	`Lami.areaForm_cyclic`	`Basic`:27	$\dim 2$, oriented; $F_1 + F_2 + F_3 = 0$
$\omega(x, y) = \lVert x\rVert\lVert y\rVert \sin \measuredangle(x, y)$	`Lami.areaForm_eq_norm_mul_norm_mul_sin_oangle`	`Basic`:36	$\dim 2$, oriented; none (all x, y)
Lami, oriented product form	`Lami.lami_oriented_mul`	`Basic`:49	$\dim 2$, oriented; $F_1 + F_2 + F_3 = 0$
Lami, oriented ratio form	`Lami.lami_oriented`	`Basic`:100	$\dim 2$, oriented; $F_1 + F_2 + F_3 = 0$; $F_1, F_2, F_3 \neq 0$
Lami for forces in $\mathbb{C}$	`Lami.lami_complex`	`Basic`:111	$V = \mathbb{C}$; $F_1 + F_2 + F_3 = 0$; $F_1, F_2, F_3 \neq 0$
Lami, unoriented product form	`Lami.lami_mul`	`Basic`:134	any dimension; $F_1 + F_2 + F_3 = 0$
Lami, unoriented ratio form	`Lami.lami`	`Basic`:148	any dimension; $F_1 + F_2 + F_3 = 0$; $F_1, F_2, F_3 \neq 0$

Informal statement	Lean name	Source	Setting / hypotheses
Lami with nonzero sines (Thm. 6)	`Lami.lami_nondegenerate`	`Basic`:180	any dimension; $F_1 + F_2 + F_3 = 0$; $F_1, F_2, F_3 \neq 0$; $\angle(F_1, F_2) \neq 0, \pi$
Parallel case: all sines 0	`Lami.lami_degenerate`	`Basic`:189	any dimension; $F_1 + F_2 + F_3 = 0$; $F_1, F_2, F_3 \neq 0$; $\angle(F_1, F_2) \in \{0, \pi\}$
Division-free form	`Lami.lami_cross`	`Basic`:211	any dimension; $F_1 + F_2 + F_3 = 0$; $F_3 \neq 0$
Converse, area-form version	`Lami.equilibrium_of_areaForm`	`Basic`:247	dim 2, oriented; $\omega(F_1, F_2) = \omega(F_2, F_3) = \omega(F_3, F_1)$; $\omega(F_1, F_2) \neq 0$
Converse, oriented ratio form	`Lami.lami_oriented_converse`	`Basic`:262	dim 2, oriented; $F_1, F_2, F_3 \neq 0$; three oriented sines $\neq 0$; the two ratio equalities

2 The method

All five proofs follow the same three steps.

1. *Coordinates.* Points are vectors of K^3 (homogeneous coordinates, Desargues), pairs in $K \times K$ (Monge), complex numbers (Simson, Napoleon), or vectors of a real inner product space (Lami). Lines, feet, apexes and centres of similitude are given by explicit formulas in these coordinates.

2. *Certificate.* The conclusion, after clearing denominators that the hypotheses make nonzero, is an identity in the coordinates. Either it holds identically (Monge), or it is an explicit multiple of a hypothesis polynomial (Napoleon, the concyclicity step of Simson), or it factors so that one factor is forced to vanish by a hypothesis (Desargues, Simson). The identity is closed by `ring` (after `field_simp` where there are denominators) or by `linear_combination` with the multiplier written in the source.

3. *Translation.* The coordinate conclusion is turned into a standard Mathlib notion: `Projectivization.IsCollinear` for Desargues, `Collinear` for Simson, `dist` for Napoleon, and norms with `Orientation.oangle` or `InnerProductGeometry.angle` for Lami. Monge's theorem itself stops at the coordinate determinant; the bridge `Monge.det3_eq_zero_iff_collinear` identifies it with `Collinear`, but the composed statement is not stated as a declaration in the main files (Section 11); the Layer-14 file of version 2 states it (`Monge.l14_monge`, Section 10.1).

The certificates are small. The largest is Desargues's, of degree 12 in eighteen variables; it takes about nine seconds to check. None of the proofs uses `decide`, `native_decide` or `polyrith`; every multiplier given to `linear_combination` is written in the source.

The interest of the method is not that it is new (it is not; see [6]) but that it makes the role of each hypothesis visible. A hypothesis is used in exactly one of three ways: to make a denominator nonzero so that `field_simp` can clear it; to supply the polynomial on the right of the certificate; or to make a translation lemma apply. Table 2 at the end of Section 7 records which is which.

3 Desargues's theorem

Representation. Let K be a field. A point of $\mathbb{P}^2(K)$ is represented by a nonzero $v \in K^3$, and Mathlib's `Projectivization.mk K v hv` is the point it represents. The bracket `Desargues.br` $u\,v\,w = u \cdot (v \times w)$ is the determinant with rows u, v, w (`Desargues.br_eq_det`, from Mathlib's `triple_product_eq_det`). If nonzero u, v represent *distinct points* of $\mathbb{P}^2(K)$, that is, u and v are linearly independent, then $u \times v \neq 0$ represents the line through them. Distinctness of the representing vectors is not enough: over $\mathbb{Q}$, $u = (1,0,0)$ and $v = (2,0,0)$ satisfy $u \neq v$ but represent the same point, and $u \times v = 0$. (In characteristic 2 the vector v is zero and represents no point at all.) Dually, if l, m represent distinct lines then $l \times m \neq 0$ represents their intersection. For nonzero o, a, a' the condition $[o\,a\,a'] = 0$ says that the three points represented by o, a, a' are collinear. When a and a' represent distinct points this is the statement "O lies on the line AA'"; when they represent the same point, the line AA' is not defined and $[o\,a\,a'] = 0$ holds for every o. All perspective hypotheses below are stated as $[o\,a\,a'] = 0$, which is meaningful in both cases.

The certificate. For all a, b, c, a', b', c' in R^3, R any commutative ring,

$$[(a \times b) \times (a' \times b'),\ (b \times c) \times (b' \times c'),\ (c \times a) \times (c' \times a')] = [abc]\,[a'b'c']\,[a \times a',\ b \times b',\ c \times c']. \quad (1)$$

This is `Desargues.desargues_det_identity`. The proof rewrites each double cross product with Mathlib's `cross_cross_eq_smul_sub_smul` ($(u \times v) \times w = (u \cdot w)v - (v \cdot w)u$), expands into coordinates, and closes the result with `ring`. No hypothesis appears: the identity is unconditional. The three points on the left are the Desargues points $P = AB \cap A'B'$, $Q = BC \cap B'C'$, $R = CA \cap C'A'$, and the last factor on the right is the bracket of the three joining lines AA', BB', CC'.

From the certificate to the theorem. If a nonzero o satisfies $[o\,a\,a'] = [o\,b\,b'] = [o\,c\,c'] = 0$, then o is orthogonal to the three vectors $a \times a'$, $b \times b'$, $c \times c'$, which are therefore linearly dependent, so their bracket vanishes (`Desargues.br_eq_zero_of_dotProduct_eq_zero`, via Mathlib's `Matrix.exists_mulVec_eq_zero_iff`). By (1), $[PQR] = 0$.

Theorem 1 (Desargues, determinant form)**.** *Let K be a field and $o, a, b, c, a', b', c' \in K^3$ with $o \neq 0$. If $[o\,a\,a'] = [o\,b\,b'] = [o\,c\,c'] = 0$, then $[PQR] = 0$, where P, Q, R are the cross-product expressions on the left of* (1)*.*

```
theorem desargues_det {o a b c a' b' c' : Fin 3 → K} (ho : o ≠ 0)
    (hOa : br o a a' = 0) (hOb : br o b b' = 0) (hOc : br o c c' = 0) :
    br ((a ×₃ b) ×₃ (a' ×₃ b')) ((b ×₃ c) ×₃ (b' ×₃ c')) ((c ×₃ a) ×₃ (c' ×₃ a')) = 0
```

The only non-degeneracy hypothesis is $o \neq 0$, and it is needed: with $o = 0$ the perspective hypotheses are vacuous and the statement would claim that $[PQR] = 0$ for all pairs of triangles, which is false (the project's negative control found 2,946 of 2,985 random non-perspective configurations over $\mathbb{Q}$ with $[PQR] \neq 0$). Everything else is allowed: coincident vertices, degenerate triangles, coincident corresponding sides. In those cases one of P, Q, R may be the zero vector and the conclusion says nothing geometric, but it is still true.

Theorem 2 (Desargues in $\mathbb{P}^2(K)$)**.** *Let K be a field and let A, B, C, A', B', C', O be points of $\mathbb{P}^2(K)$. Suppose $A \neq B$, $B \neq C$, $C \neq A$, $A' \neq B'$, $B' \neq C'$, $C' \neq A'$, and $AB \neq A'B'$, $BC \neq B'C'$, $CA \neq C'A'$ as lines. Let $o, a, \ldots, c'$ be nonzero representatives of $O, A, \ldots, C'$, and suppose $[o\,a\,a'] = [o\,b\,b'] = [o\,c\,c'] = 0$ (that is, O, A, A' are collinear, and likewise for the other two pairs; when $A \neq A'$ this says that O lies on AA'). Then the points $AB \cap A'B'$, $BC \cap B'C'$ and $CA \cap C'A'$ are collinear.*

This is `Desargues.desargues_mk`. Points are written `mk K a ha` and so on; the line through two points and the intersection of two lines are both Mathlib's `Projectivization.cross`; and the conclusion is `Projectivization.IsCollinear` of the three-point set. The perspective hypotheses are stated on representatives (`br o a a' = 0`); no hypothesis $A \neq A'$ is imposed, and a coincidence $A = A'$ simply makes the corresponding perspective hypothesis vacuous. The six distinctness hypotheses on vertices and the three on sides have a precise role. Mathlib's `Projectivization.cross` returns its first argument when the cross product of representatives vanishes, so it computes the line through X and Y only when $X \neq Y$, and the intersection of two lines only when they are distinct (`Projectivization.cross_mk_of_ne`). The nine hypotheses are exactly what is needed to rewrite every `cross` in the statement as the cross product of representatives, after which Theorem 1 and `Desargues.isCollinear_mk_of_br_eq_zero` finish the proof. They are not needed for (1). No hypothesis is placed on O beyond its being a point; in particular O may coincide with a vertex, and the triangles need not be non-degenerate. No assumption on the characteristic of K is needed.

The converse. If both triangles are non-degenerate, $[abc] \neq 0 \neq [a'b'c']$, then by (1) $[PQR] = 0$ forces $[AA'\,BB'\,CC'] = 0$ (`Desargues.desargues_converse_det`). Non-degeneracy is necessary for this argument: if either triangle is degenerate, the right-hand side of (1) vanishes for every configuration and gives no information. The projective form `Desargues.desargues_converse_mk` assumes that the joining vectors $a \times a'$, $b \times b'$, $c \times c'$ and the Desargues points P, Q, R are nonzero, and concludes that the three lines AA', BB', CC', regarded as points of the dual plane, are collinear. That this is concurrency of the three lines in the primal plane is the content of `Desargues.exists_dotProduct_eq_zero_of_br_eq_zero`: a vanishing bracket gives a nonzero o orthogonal to all three, that is, a point on all three lines. A reader looking for a statement of the form "there is a point O on AA', BB' and CC'" has to compose the two declarations; Mathlib has no concurrency predicate. Version 2 does this composition in the Layer-15 file: `Desargues.Classical.cov_A1_b` gives a point on the three lines, with lines as spans in Mathlib's projective plane (Section 10.1).

Non-vacuity. `Examples.lean` works out the configuration $O = (0{:}0{:}1)$, $A = (1{:}0{:}0)$, $B = (0{:}1{:}0)$, $C = (1{:}1{:}1)$, $A' = (2{:}0{:}3)$, $B' = (0{:}5{:}{-1})$, $C' = (4{:}4{:}11)$ over $\mathbb{Q}$. Both triangles are non-degenerate (`Desargues.ex_nonvacuous`), the triangles are in perspective from O (`Desargues.ex_perspective`), the Desargues points are $P = (-2{:}-15{:}0)$, $Q = (-4{:}-39{:}-4)$, $R = (2{:}-12{:}-12)$, all nonzero (`Desargues.ex_points_ne_zero`), and $[PQR] = 0$ is obtained both from the theorem (`Desargues.ex_desargues`) and by direct computation (`Desargues.ex_desargues_compute`).

4 Monge's theorem

For circles with centres $O_1, O_2 \in K^2$ and radii $r_1 \neq r_2$, the external centre of similitude is the point dividing O_1O_2 externally in the ratio $r_1 : r_2$,

$$S_{12} = \frac{r_1 O_2 - r_2 O_1}{r_1 - r_2},$$

which is `Monge.ext` $\mathtt{O}_1$ $\mathtt{O}_2$ $\mathtt{r}_1$ $\mathtt{r}_2$, defined componentwise. Collinearity is the vanishing of `Monge.det3` $A\,B\,C = (B-A)\times(C-A)$, the signed double area; over every field, $\det_3(A,B,C) = 0$ is equivalent to Mathlib's `Collinear K {A, B, C}` in $K \times K$ (`Monge.det3_eq_zero_iff_collinear`).

Theorem 3 (Monge)**.** *Let K be a field, $O_1, O_2, O_3 \in K^2$, and $r_1, r_2, r_3 \in K$ pairwise distinct. Then* $\det_3(S_{12}, S_{23}, S_{31}) = 0$.

```
theorem monge (O₁ O₂ O₃ : K × K) (r₁ r₂ r₃ : K)
    (h12 : r₁ ≠ r₂) (h23 : r₂ ≠ r₃) (h31 : r₃ ≠ r₁) :
    det3 (ext O₁ O₂ r₁ r₂) (ext O₂ O₃ r₂ r₃) (ext O₃ O₁ r₃ r₁) = 0
```

The certificate. After multiplying by $(r_1-r_2)(r_2-r_3)(r_3-r_1)$ the determinant is a polynomial in the nine variables that vanishes identically. The proof is `unfold det3 ext; field_simp; ring`. There is no hypothesis polynomial on the right: the three hypotheses are used only to make the three denominators nonzero.

Hypotheses. *Scope of the geometric discussion.* Everything said below about circles, centres of similitude, homotheties and tangents concerns genuine circles, with $K = \mathbb{R}$ and positive real radii. It is not valid for every field element that `Monge.monge` admits. For instance, $r_1 = 1$, $r_2 = 2$, $r_3 = 0$ satisfies the three hypotheses, but then (as for any $r_1, r_2 \neq 0$ with $r_3 = 0$) $S_{23} = S_{31} = O_3$, so two of the three points coincide, the line $S_{23}S_{31}$ is undefined, and the conclusion $\det_3 = 0$ holds trivially. More generally, if $r_i = 0$ the homothety ratio r_j/r_i displayed below is undefined classically (Lean would read it as 0). The Lean theorem is an identity over an arbitrary field; its geometric reading is restricted to $r_1, r_2, r_3 > 0$ pairwise distinct.

Distinct radii are exactly the condition that each S_{ij} is a finite point. They cannot simply be dropped from the Lean statement: Lean defines $x/0 = 0$, so with $r_1 = r_2$ the term `ext` $\mathtt{O}_1$ $\mathtt{O}_2$ $\mathtt{r}_1$ $\mathtt{r}_2$ is the origin, and the determinant need not vanish. For example, with centres $(0,0), (4,0), (0,3)$ and radii $1, 1, 3$, a computation in exact rational arithmetic with Lean's division convention gives $\det_3 = -9$. (This computation was done in Python, not in Lean.) Geometrically, the equal-radius cases are of several kinds, none of which is formalised. If exactly two radii agree, say $r_1 = r_2 \neq r_3$, and $O_1 \neq O_2$, then S_{12} is the point at infinity in the direction of O_1O_2, and the projective form of Monge's theorem says that $S_{23}S_{31}$ is parallel to O_1O_2 (a homothety centred at O_3 maps O_1, O_2 to S_{31}, S_{23}). If all three radii agree and the centres are distinct, then *all three* centres of similitude are at infinity and the statement degenerates to the triviality that three points of the line at infinity are collinear; there is no finite Monge line. If two circles are identical ($O_i = O_j$, $r_i = r_j$), every point of the plane is fixed by the identity map and there is no distinguished centre of similitude at all. No hypothesis on the centres is needed for the *identity*: it holds for all O_i, including coincident centres. For $r_i \neq r_j$ and $O_i = O_j$ (concentric circles) the formula gives $S_{ij} = O_i$, which is indeed the centre of the homothety $x \mapsto O_i + \frac{r_j}{r_i}(x - O_i)$, but the circles have no common tangents. More generally, the formula is meaningful without any geometric configuration assumption only as the centre of the positive-ratio homothety; its classical description as the meeting point of the two external common tangents requires that those tangents exist, which fails when one circle lies strictly inside the other. Nor are the radii required to be positive in the formal statement. Because `Monge.monge` is universally quantified over $r_1, r_2, r_3 \in K$, replacing r_2 by $-r_2$ turns S_{12} and S_{23} into the *internal* centres of similitude, and gives the classical variant "two internal centres and the third external centre are collinear" under the hypotheses $r_1 \neq -r_2$, $-r_2 \neq r_3$, $r_3 \neq r_1$, where now $r_1, r_2, r_3 > 0$ are the actual radii. We note this only as an instantiation; it is not stated as a separate declaration in the main files. Version 2 proves the companion form (one external and two internal centres,

for circles given as Mathlib spheres) by the same sign-change argument, using $r_3 \mapsto -r_3$ to obtain E_{12}, I_{23}, I_{13}, as `Monge.Classical.cov_A2_b` (Section 10.1).

Non-vacuity. For centres $(0,0),(4,0),(0,3)$ and radii $1,2,3$ over $\mathbb{Q}$, the external centres are $(-4,0)$, $(12,-6)$ and $(0,-3/2)$ (`Monge.monge_example_values`), and $\det_3 = 0$ (`Monge.monge_example`).

Scope. Neither circles nor tangent lines appear in the formal statement. The radii are elements of K and `Monge.ext` is a formula. The bridge `Monge.ext_eq_lineMap_homothety` shows, for $r_1 \neq 0$ and $r_1 \neq r_2$, that S_{12} is the point `AffineMap.lineMap O₁ O₂ (r₁ / (r₁ - r₂))` of the line O_1O_2 and that the homothety (`AffineMap.homothety`) with centre S_{12} and ratio r_2/r_1 sends O_1 to O_2. The hypothesis $r_1 \neq 0$ is needed in addition to $r_1 \neq r_2$: for $r_1 = 0$ the formula gives $S_{12} = O_1$, which is not a centre of similitude, and the ratio r_2/r_1 is undefined (Lean reads it as 0). The lemma is about centres only; in the main files, that this homothety maps the one circle onto the other is not formalised. Version 2 proves it in the Layer-15 file: for two circles with distinct centres and positive unequal radii given as Mathlib spheres, a point at which the homothety of ratio r_t/r_s maps the one circle onto the other exists and is unique (`Monge.Classical.cov_A2_obligation_ext`), it is the point `Monge.ext` (`Monge.Classical.ext_eq_of_isExt`), and for three circles with pairwise distinct radii the three such points are `Collinear` (`Monge.Classical.cov_A2_a`). Not formalised is the formulation in which the external common tangents of each pair meet at S_{ij} (a formulation that applies only when those tangents exist, so not to nested circles). Mathlib's `Geometry/Euclidean/MongePoint.lean` concerns the Monge point of a simplex, a different theorem, and is not used.

Prior formalisations. This theorem is not new to Lean, and our proof is not the first; two earlier complete Lean proofs exist (Section 9).

5 The Simson line

Representation. The Euclidean plane is $\mathbb{C}$, regarded as a real affine space. For $u \neq v$ the foot of the perpendicular from p to the line uv is `Simson.perpFoot` $p\,u\,v = u + t(v-u)$ with $t = \langle p-u, v-u\rangle / |v-u|^2$, written in real and imaginary parts (`Simson.footParam`). Two lemmas say that this is the foot: it lies on Mathlib's `line[ℝ, u, v]` (`Simson.perpFoot_mem_line`), and $p -$ foot is orthogonal to $v-u$ when $u \neq v$ (`Simson.inner_perpFoot`, stated in coordinates). Moreover the parameter is $t = \langle p-u, v-u\rangle / \|v-u\|^2$ for Mathlib's real inner product on $\mathbb{C}$ (`Simson.footParam_eq_inner`), and the foot is Mathlib's `EuclideanGeometry.orthogonalProjection` of p onto `line[ℝ, u, v]` (`Simson.perpFoot_eq_orthogonalProjection`), with no hypothesis (for $u = v$ both sides are u). The signed double area is `Simson.area2`, and `Simson.concyc` $a\,b\,c\,p$ is the 4×4 determinant with rows $(x, y, x^2+y^2, 1)$, written out by cofactor expansion. For a non-degenerate triangle, area2$(a,b,c) \neq 0$, it vanishes exactly when $\{a,b,c,p\}$ is Mathlib's `EuclideanGeometry.Cospherical` (`Simson.concyc_eq_zero_iff_cospherical`). The hypothesis cannot be dropped, and concyc does not vanish merely because a,b,c are collinear: for $a,b,c = (0,0),(1,0),(2,0)$ and $p = (0,1)$ one has concyc$(a,b,c,p) = 2$, while four points on one line have concyc $= 0$ without being cospherical. (The comment on `Simson.concyc` in the cited source describes its zero set inaccurately; the bridge lemma, with its hypothesis, is the correct statement.)

The certificate. Write $D_1 = |c-b|^2$, $D_2 = |a-c|^2$, $D_3 = |b-a|^2$ and $S = \mathrm{area2}(a,b,c)$. For pairwise distinct a, b, c and every p,

$$\mathrm{area2}(\mathrm{foot}_{bc}(p), \mathrm{foot}_{ca}(p), \mathrm{foot}_{ab}(p)) \cdot D_1 D_2 D_3 = S^2 \cdot \mathrm{concyc}(a,b,c,p). \tag{2}$$

This is `Simson.area2_perpFoot_mul`, proved by `field_simp; ring`; the distinctness hypotheses make the D_i nonzero (`Simson.sqD_ne_zero`). A second, linear certificate shows that four points at distance r from a common o have concyc $= 0$ (`Simson.concyc_eq_zero_of_dist`): concyc is a combination of the four equations $|x-o|^2 = r^2$ with the cofactors as multipliers, given explicitly to `linear_combination`. The factorisation (2) was found with `sympy` (`code/cert.py`, `code/gen.py`, `code/gen2.py`) and then checked by Lean.

Theorem 4 (Wallace–Simson, forward direction)**.** *Let $a, b, c \in \mathbb{C}$ be pairwise distinct and let $p \in \mathbb{C}$ be such that $\{a, b, c, p\}$ is cospherical. Then the feet of the perpendiculars from p to the lines bc, ca, ab are collinear.*

```
theorem simson {a b c p : ℂ} (hab : a ≠ b) (hbc : b ≠ c) (hca : c ≠ a)
    (hcirc : EuclideanGeometry.Cospherical ({a, b, c, p} : Set ℂ)) :
    Collinear ℝ ({perpFoot p b c, perpFoot p c a, perpFoot p a b} : Set ℂ)
```

The hypothesis is Mathlib's `EuclideanGeometry.Cospherical`, which in the plane $\mathbb{C}$ means that the four points lie on a common circle (of any radius, including radius 0, which the distinctness hypotheses exclude). The conclusion is Mathlib's `Collinear` $\mathbb{R}$. The intermediate `Simson.simson_area` states the conclusion as area2 $= 0$ with the circle given by a centre o and radius r.

Hypotheses. The only hypotheses are distinctness of the three vertices and the circle condition. Distinctness is used only to make the feet well defined and the D_i nonzero. No separate hypothesis that the triangle is non-degenerate is needed: three distinct points on a circle are not collinear, and in any case (2) does not need it. The point p may be a vertex, and right-angled triangles are included; the identity covers all of these.

The converse. The main files of our development prove only the forward direction of the Wallace–Simson theorem. For a non-degenerate triangle ($S \neq 0$, which implies distinctness), (2) gives area2(feet) $= 0 \iff$ concyc$(a,b,c,p) = 0$ (`Simson.area2_perpFoot_eq_zero_iff`). This is an equivalence between two *local determinant expressions*; it is not the geometric converse. Neither side is a Mathlib notion. On the circle side the bridge is now two-way for $S \neq 0$ (`Simson.concyc_eq_zero_iff_cospherical`), but on the feet side it is one-way: `Simson.collinear_of_area2_eq_zero` turns area2 $= 0$ into `Collinear`, and there is no lemma from `Collinear` back to area2 $= 0$. The geometric converse, "if the feet are collinear then p lies on the circumcircle", was therefore not formalised in the first version of this paper; the missing step was that converse bridge for area2. Version 2 supplies it in the Layer-15 file: `Simson.Classical.area2_eq_zero_of_collinear` (eight lines) proves that three collinear points have area2 $= 0$, and `Simson.Classical.cov_A3_b` chains it with the two lemmas above to prove the converse for a Mathlib `Affine.Triangle` in `EuclideanSpace ℝ (Fin 2)`, with conclusion membership in Mathlib's `Affine.Simplex.circumsphere` (Section 10.1). The converse was proved earlier in the Isabelle AFP entry and in Guilhot's Rocq development cited in Section 9.

Non-vacuity. The library contains an `example` applying `Simson.simson` to the triangle $1, i, -1$ on the unit circle with $p = -i$, and a negative control showing that for $p = 0$ (the centre) the pedal triangle has nonzero area.

Where the reals enter. The certificate (2) is an identity of polynomials in the eight real coordinates (x, y) of a, b, c, p and, as a polynomial identity, it holds over any commutative ring. The proof uses the reals in two places: the foot is defined through the Euclidean inner product, and $D_i \neq 0$ for $a \neq b$ uses that a sum of two real squares vanishes only if both do. The second step is what fails if the coordinates are allowed to range over a field such as $\mathbb{C}$ (or any field containing a square root of -1) while D_i is kept as the *bilinear* expression $(\Delta x)^2 + (\Delta y)^2$, which is what the polynomial identity contains: this quadratic form is isotropic there, e.g. $\Delta = (1, i)$ gives $1^2 + i^2 = 0$ although $\Delta \neq 0$, so two distinct points can have $D = 0$ and the foot is then undefined. (The Hermitian expression $|\Delta x|^2 + |\Delta y|^2$ is positive for $\Delta \neq 0$, but it is not a polynomial in the coordinates and does not appear in (2).) Over an arbitrary field one would have to assume the side vectors non-isotropic. This is why the theorem is stated in $\mathbb{C}$ regarded as the real Euclidean plane, with real coordinates, and not over an arbitrary field.

6 Napoleon's theorem

Representation. The plane is $\mathbb{C}$. For $\omega \in \mathbb{C}$, `Napoleon.apex` $\omega\, P\, Q = P + \omega(Q - P)$ is the image of Q under the map $z \mapsto P + \omega(z - P)$. For general ω this is a spiral similarity (or, for $\omega = 0$, constant), not a rotation; it is a rotation about P exactly when $|\omega| = 1$. The two rotations are $\omega_o = \frac{1}{2} - \frac{\sqrt{3}}{2} i$ (by $-\pi/3$, `Napoleon.ÏĽo`) and $\omega_i = \overline{\omega_o}$ (`Napoleon.ÏĽi`). The Napoleon points G_a, G_b, G_c are the centroids of the triangles erected on BC, CA, AB (`Napoleon.Ga` and its two companions).

The certificate. If $\omega^2 - \omega + 1 = 0$, then for all A, B, C

$$G_b - G_a - \omega(G_c - G_a) = \tfrac{1}{3}(A - 2B + C)\,(\omega^2 - \omega + 1). \tag{3}$$

This is `Napoleon.key`, proved by `linear_combination (A - 2 * B + C) / 3 * h`. It says that G_b is the image of G_c under $z \mapsto G_a + \omega(z - G_a)$. Over $\mathbb{C}$ the hypothesis $\omega^2 - \omega + 1 = 0$ already forces $\omega = e^{\pm i\pi/3}$, so this map is in fact a rotation by $\pm\pi/3$; but `Napoleon.key` is a ring identity and does not use that.

More precisely, (3) needs no hypothesis at all: both sides are polynomials in A, B, C, ω, and the identity holds for every ω, not merely for roots of $\omega^2 - \omega + 1$. The Lean declaration `Napoleon.key` states the specialisation $G_b - G_a = \omega(G_c - G_a)$ that is used below, and carries $\omega^2 - \omega + 1 = 0$ as a hypothesis; as formalised it is therefore marginally weaker than the displayed identity, which we have kept in the conditional form in which the geometry consumes it.

Theorem 5 (Napoleon)**.** *For all $A, B, C \in \mathbb{C}$, the centres G_a, G_b, G_c of the equilateral triangles erected on the sides by the rotation ω_o satisfy $|G_aG_b| = |G_bG_c| = |G_cG_a|$; and the same holds for the rotation ω_i.*

```
theorem napoleon_outer (A B C : ℂ) :
    dist (Ga ωo A B C) (Gb ωo A B C) = dist (Gb ωo A B C) (Gc ωo A B C) ∧
    dist (Gb ωo A B C) (Gc ωo A B C) = dist (Gc ωo A B C) (Ga ωo A B C)
```

Both statements (`Napoleon.napoleon_outer`, `Napoleon.napoleon_inner`) are instances of `Napoleon.napoleon_general`, which holds for every ω with $\omega^2 - \omega + 1 = 0$, $|\omega| = 1$ and $|1 - \omega| = 1$. These three hypotheses are not independent: the roots of $\omega^2 - \omega + 1$ are $\omega = \frac{1}{2} \pm \frac{\sqrt{3}}{2} i = e^{\pm i\pi/3}$, and both satisfy $|\omega| = |1 - \omega| = 1$, so the two norm hypotheses follow from the first. (Conversely, $|\omega| = |1 - \omega| = 1$ alone also characterises $e^{\pm i\pi/3}$.) The Lean statement keeps all three because the

proof uses them in different places: the first drives (3), and the other two turn it into equalities of distances, since $G_b - G_a = \omega(G_c - G_a)$ and $G_c - G_b = (1-\omega)(G_c - G_a)$. Deriving the norm hypotheses from the polynomial one would remove two redundant binders; we have not done so. The lemma `Napoleon.apex_equilateral` checks that the erected triangles are equilateral, under the same two norm hypotheses.

Hypotheses. Theorem 5 has none. For a degenerate triangle the conclusion still holds, possibly with a Napoleon triangle that is a single point. Orientation enters only in the interpretation of ω_o as "outward". The lemma `Napoleon.apex_outward` computes the signed area $\mathrm{signedArea}(P, Q, \mathrm{apex}\,\omega_o\,P\,Q) = -\frac{\sqrt{3}}{4}|Q-P|^2$, for all P, Q. So every triangle erected with ω_o on a base with $P \neq Q$ is clockwise; when $P = Q$ the erected "triangle" is the single point P, its signed area is 0, and it has no orientation. If ABC is counter-clockwise ($\mathrm{signedArea}(A, B, C) > 0$, which forces the vertices to be distinct), each apex therefore lies on the other side of its base from the third vertex, and the ω_o triangles are the outer ones; for a clockwise ABC the outer triangles are the ω_i ones, and `Napoleon.napoleon_inner` covers that case. "Outward" is thus expressed through signed area rather than through half-planes; in the main files the step from the signed-area statement to "on the opposite side of the line" is not formalised. Version 2 adds it: `Napoleon.l14_napoleon_outer` takes apexes defined by equal sides and Mathlib's strict opposite side (`AffineSubspace.SOppSide`), with no orientation hypothesis, and `Napoleon.Classical.cov_A4_a` proves the classical outer statement in `EuclideanSpace ℝ (Fin 2)`, including that the outer Napoleon triangle of a triangle is non-degenerate (Section 10.1). Mixing orientations (erecting some triangles with ω_o and some with ω_i) makes the statement false; the project's numerical check found a spread of 1.15 between the three distances in such a configuration.

Non-vacuity. An `example` checks that for $A = 0$, $B = 1$, $C = i$ the triangle is counter-clockwise, the Napoleon equality holds, and $|G_a G_b| \neq 0$.

Prior formalisations. This theorem is not new to Lean, and our version is not the first in a proof assistant to treat the exterior condition; see Section 9.

7 Lami's theorem

Lami's theorem says that if three forces F_1, F_2, F_3 acting at a point are in equilibrium, $F_1 + F_2 + F_3 = 0$, then

$$\frac{|F_1|}{\sin\alpha} = \frac{|F_2|}{\sin\beta} = \frac{|F_3|}{\sin\gamma},$$

where α is the angle between F_2 and F_3, β the angle between F_3 and F_1, and γ the angle between F_1 and F_2.

Oriented form and the certificate. Let V be a real inner product space of dimension two with an orientation o, and let ω be its area form (Mathlib's `Orientation.areaForm`). Substituting $F_3 = -F_1 - F_2$ and using bilinearity and $\omega(x, x) = 0$ gives

$$\omega(F_1, F_2) = \omega(F_2, F_3) = \omega(F_3, F_1) \tag{4}$$

(`Lami.areaForm_cyclic`, closed by `ring` after `simp`). This is the certificate. The translation to sines is `Lami.areaForm_eq_norm_mul_norm_mul_sin_oangle`, $\omega(x, y) = \|x\|\,\|y\| \sin \measuredangle(x, y)$ for

all x, y including zero, proved through Mathlib's Kähler form and `Complex.sin_arg`. Together they give the product form `Lami.lami_oriented_mul`, with equilibrium as the only hypothesis: $\|F_2\|\|F_3\| \sin \measuredangle(F_2, F_3) = \|F_3\|\|F_1\| \sin \measuredangle(F_3, F_1) = \|F_1\|\|F_2\| \sin \measuredangle(F_1, F_2)$. The ratio form `Lami.lami_oriented` needs $F_1, F_2, F_3 \neq 0$ and follows from the product form by `Lami.ratio_of_mul`: if all $n_i \neq 0$ and $n_2 n_3 s_1 = n_3 n_1 s_2 = n_1 n_2 s_3$, then either all s_i vanish or none does, and $n_1/s_1 = n_2/s_2 = n_3/s_3$ in Lean's convention. `Lami.lami_complex` is the instance $V = \mathbb{C}$ with its standard orientation.

Unoriented form. With Mathlib's unoriented angle `InnerProductGeometry.angle` (values in $[0, \pi]$), $\|x\|\|y\| \sin \measuredangle(x, y)$ is the square root of the Gram determinant $\langle x, x\rangle\langle y, y\rangle - \langle x, y\rangle^2$ (`Lami.norm_mul_norm_mul_sin_angle`). The three Gram determinants of (F_2, F_3), (F_3, F_1), (F_1, F_2) agree once $F_3 = -F_1 - F_2$ is substituted, again by `ring`. This gives `Lami.lami_mul` and `Lami.lami` in every real inner product space; no two-dimensionality is assumed, since the three forces span a space of dimension at most two.

Theorem 6 (Lami, non-degenerate form)**.** *Let V be a real inner product space and $F_1, F_2, F_3 \in V$ nonzero with $F_1 + F_2 + F_3 = 0$ and $\measuredangle(F_1, F_2) \notin \{0, \pi\}$. Then $\sin\alpha$, $\sin\beta$, $\sin\gamma$ are nonzero and $\|F_1\|/\sin\alpha = \|F_2\|/\sin\beta = \|F_3\|/\sin\gamma$.*

```
theorem lami_nondegenerate {F₁ F₂ F₃ : V} (h : F₁ + F₂ + F₃ = 0)
    (h₁ : F₁ ≠ 0) (h₂ : F₂ ≠ 0) (h₃ : F₃ ≠ 0)
    (hγ : angle F₁ F₂ ≠ 0) (hγ' : angle F₁ F₂ ≠ π) :
    (Real.sin (angle F₂ F₃) ≠ 0 ∧ Real.sin (angle F₃ F₁) ≠ 0 ∧ Real.sin (angle F₁ F₂) ≠
    0) ∧
      ‖F₁‖ / Real.sin (angle F₂ F₃) = ‖F₂‖ / Real.sin (angle F₃ F₁) ∧
      ‖F₂‖ / Real.sin (angle F₃ F₁) = ‖F₃‖ / Real.sin (angle F₁ F₂)
```

This is the declaration to cite for the classical theorem. The statement `Lami.lami` has the same conclusion without the sine conjunct and without the non-parallel hypotheses; it is true, but in the parallel case it is true only because Lean defines $x/0 = 0$. `Lami.lami_degenerate` makes this explicit: if $F_1 \parallel F_2$ then all three sines vanish. The division-free form `Lami.lami_cross`, $\|F_1\| \sin\beta = \|F_2\| \sin\alpha$, needs only $F_3 \neq 0$.

Converse. In dimension two, if the three cyclic area forms are equal and nonzero, the forces are in equilibrium (`Lami.equilibrium_of_areaForm`, via Mathlib's `Orientation.inner_mul_areaForm_sub`). Hence three nonzero forces with nonzero oriented sines whose magnitudes are proportional to the sines of the opposite oriented angles are in equilibrium (`Lami.lami_oriented_converse`). Orientation is essential here: with unoriented angles the converse is false, because replacing F_1 by $-F_1$ leaves every unoriented sine unchanged but destroys equilibrium.

The law of sines. Lami's theorem is the law of sines for the force triangle with sides $\|F_1\|, \|F_2\|, \|F_3\|$ and interior angles $\pi - \alpha, \pi - \beta, \pi - \gamma$. The declaration `Lami.lami_via_law_sin` only restates Mathlib's `InnerProductGeometry.sin_angle_mul_norm_eq_sin_angle_mul_norm`. It is not used in the proofs, and the connection between Lami's theorem and the force triangle is not formalised.

8 Does the statement mean what it says?

Each development introduces some local notions. A reader has to check that they mean what their names suggest. The lemmas available for this are as follows.

- *Desargues.* The bracket is the determinant (`Desargues.br_eq_det`); a vanishing bracket is equivalent to Mathlib's `Projectivization.IsCollinear` of the three points (`Desargues.isCollinear_mk_of_br_eq_zero`, `Desargues.br_eq_zero_of_isCollinear_mk`); and a vanishing bracket of three lines is concurrency (`Desargues.exists_dotProduct_eq_zero_of_br_eq_zero`, `Desargues.br_eq_zero_of_dotProduct_eq_zero`). Lines and intersections in Theorem 2 are Mathlib's own. For the worked example, the representatives are nonzero (`Desargues.ex_vertices_ne_zero`), the centre O is Mathlib's intersection `Projectivization.cross` of the lines AA' and BB' (`Desargues.exO_eq_cross`), and each of A', B', C' lies on the line from O to the corresponding vertex in the sense of `Projectivization.IsCollinear` (`Desargues.exA'_isCollinear`, `Desargues.exB'_isCollinear`, `Desargues.exC'_isCollinear`).

- *Monge.* `Monge.det3` $A\,B\,C = 0$ is equivalent to Mathlib's `Collinear` over every field (`Monge.det3_eq_zero_iff_collinear`). For $r_1 \neq 0$ and $r_1 \neq r_2$, `Monge.ext` lies on the line O_1O_2 (`AffineMap.lineMap`) and is the centre of the homothety (`AffineMap.homothety`) of ratio r_2/r_1 sending O_1 to O_2 (`Monge.ext_eq_lineMap_homothety`); for $r_1 = 0$ it is not a centre of similitude (Section 4).

- *Simson.* `Simson.perpFoot` is Mathlib's `EuclideanGeometry.orthogonalProjection` onto the line (`Simson.perpFoot_eq_orthogonalProjection`), and `Simson.footParam` is the usual inner-product coefficient (`Simson.footParam_eq_inner`). `Simson.area2` has a one-way bridge in the main files, `Simson.collinear_of_area2_eq_zero` (area2 $= 0$ implies `Collinear`), but no lemma in the other direction there; version 2 adds it in the Layer-15 file (`Simson.Classical.area2_eq_zero_of_collinear`). `Simson.concyc` is equivalent to `Cospherical` under the hypothesis area2$(a, b, c) \neq 0$ (`Simson.concyc_eq_zero_iff_cospherical`), which cannot be dropped (Section 5). The hypothesis and the conclusion of Theorem 4 are Mathlib notions.

- *Napoleon.* The conclusion uses Mathlib's `dist`. `Napoleon.apex_equilateral` shows that the erected triangles are equilateral, and `Napoleon.apex_outward` fixes their orientation (for bases with $P \neq Q$). Each local definition has a bridge: `Napoleon.centroid` is Mathlib's `Finset.centroid` (`Napoleon.centroid_eq_finset_centroid`), and so are the Napoleon points (`Napoleon.Ga_eq_centroid`, `Napoleon.Gb_eq_centroid`, `Napoleon.Gc_eq_centroid`); for a unit complex number, `Napoleon.apex` is Mathlib's `rotation` about P (`Napoleon.apex_eq_rotation`); `Napoleon.signedArea` is half of `Complex.orientation`'s area form (`Napoleon.signedArea_eq_areaForm`); and $\omega_o = e^{-i\pi/3}$, $\omega_i = e^{i\pi/3}$ (`Napoleon.ÏĽo_eq_exp`, `Napoleon.ÏĽi_eq_exp`).

- *Lami.* Hypotheses and conclusions use only norms and Mathlib's oriented and unoriented angles. `Lami.areaForm_eq_norm_mul_norm_mul_sin_oangle` and `Lami.norm_mul_norm_mul_sin_angle` relate the algebraic objects of the proofs to those angles.

The house bridge inventory (Section 10) lists every local definition and whether it is bridged.

9 Related work

Each project began with a prior-art search of Mathlib (sources and the lists `docs/100.yaml` and `docs/1000.yaml`), GitHub code, repository and pull-request search, the Isabelle Archive of Formal Proofs [1], and the web. The searches were made between 30 September and 3 October 2026. They are recorded in the `SPIKE_REPORT.md` of each repository; we summarise them here.

Desargues. Desargues's theorem is number 87 on Wiedijk's list of 100 theorems [19], which records formalisations in HOL Light (Harrison), Isabelle (Bordg), Rocq (Guilhot and Narboux), Metamath (Megill) and Mizar (Leończuk and Prażmowski). The HOL Light proof [11] is a close precedent for ours: it builds the real projective plane from nonzero vectors of $\mathbb{R}^3$, proves that collinearity of three points is the vanishing of their bracket (`COLLINEAR_BRACKET`), and proves Desargues's theorem (`DESARGUES_DIRECT`, following Richter-Gebert's bracket proof) by reducing it to bracket identities closed by the ring decision procedure `REAL_RING`, under seventeen non-collinearity hypotheses. It is thus also a coordinate certificate proof, over $\mathbb{R}$ rather than over an arbitrary field, and with a different factorisation from (1). In Isabelle, the AFP entry *Projective Geometry* [2] proves the theorem from an axiomatisation of projective space via matroid rank; Magaud, Narboux and Schreck give a Coq proof from incidence axioms [13]; these two are synthetic. In Mathlib, `docs/100.yaml` (at the pinned revision) lists the theorem with no declaration, and `Projectivization/Constructions.lean` supplies the cross-product API that we use but no Desargues statement. The open, unmerged Mathlib pull request #37456 [15] (created 31 March 2026; head commit `0900538c`) proves the affine *parallel* form of Desargues's theorem in an affine space. We therefore do not claim a first Lean proof of Desargues's theorem: we found no Lean proof of the *projective-plane* formulation, while an unmerged pull request covers the affine (parallel) form.

Monge. Monge's three-circle theorem is *not* new to Lean, and we withdraw the claim, made in an earlier draft of this paper, that ours is the first Lean proof. Two independent, complete Lean proofs exist. (i) The benchmark repository `epfl-lara/icml-26-lean-challenges` [17], file `autformalization/problems/geometry/L2/geo_gen_L2_005/ShadowBench/Source/Main.lean` at commit `7541fcecef97cc124f38355a2cd2ba0ec8485dd5` (12 June 2026; 135 lines, no `sorry`), proves `EuclideanGeometry.monges_circle_theorem` (line 73), with the external homothety centres given explicitly. This predates our development (3 October 2026) by about four months, and on it alone the claim of priority is withdrawn. (ii) The repository `facebookresearch/atlas-lean` [3], file `MathlibExt/Geometry/Euclidean/MongeCircles.lean`, added by pull request #62 at commit `39febc62dfe68f9651c72eedf7916a7a070141a8` (5 October 2026, 21:40 UTC; 83 lines, no `sorry`), proves `monge_three_circles_general`, stated in an arbitrary real module with distinct weights as the only hypothesis, and `monge_three_circles`. Our development was made independently of it. *Dates and what they show.* Our Monge development dates from 3 October 2026 and was made public at `github.com/jpbald93/monge` on 5 October 2026 (repository created 17:16 UTC). The Atlas files `MongeCircles.lean` and `NapoleonTheorem.lean` first became public on 5 October 2026 between 21:37 and 21:40 UTC, when a batch of import pull requests (#46–#65) was opened; this is after our work. These are dates of *public* availability only. The Atlas pull requests describe the files as migrated from an upstream repository, `fairinternal/formal-math`, which is private, so we cannot see when the Atlas files were written, and earlier private authorship cannot be excluded. Likewise, our own date of 3 October 2026 is a private date that readers cannot check; only the public dates are verifiable. We therefore claim no priority over the Atlas proofs, only that ours was not derived from them. The ShadowBench proof (12 June 2026) is earlier on any reading. Our searches (GitHub code search for "monge" in Lean, Mathlib's `docs/1000.yaml`, which lists

the theorem with no declaration, Mathlib pull requests, the AFP, HOL Light, Coq and Mizar) missed the first of these; they were identified during revision. What remains distinguishable in our version is narrow: the statement is in coordinates $K \times K$ over an arbitrary field K, including fields of positive characteristic, rather than over $\mathbb{R}$ or a real module, and Section 4 analyses which hypotheses are needed and why (pairwise distinct radii only, none on the centres, and the meaning of zero, negative or equal radii). We do not claim that our hypotheses are weaker than those of the Atlas proof, whose general form also assumes only distinct weights. We did not find a formalisation of the three-circle theorem in other systems, but given the above we do not regard that search as reliable.

Simson. The Wallace–Simson theorem is *not* new to formal mathematics, and the first version of ours was weaker than the existing ones in one respect. The Isabelle AFP entry *The Wallace–Simson Line Theorem in Isabelle/HOL* by Ramos, Hulak and de Queiroz [5] (session `Simson`, 2026) uses complex coordinates, as we do, and proves both the theorem and its geometric converse: if the three feet are collinear, then the point lies on the circumcircle. Guilhot's Rocq development of oriented angles [10, 9] proves the theorem `droite_Simson` in file `point_Simson.v` as an equivalence between collinearity of the feet and concyclicity, that is, with its converse, under the additional hypothesis `~ colineaire (vec C A) (vec C M)` (the point M is not on line CA). Possessing the converse does not make its admissible-configuration domain uniformly stronger than ours: our forward theorem excludes no position of the point on the circle; it assumes only distinct vertices and the circle condition. HOL Light's `Complex/grobner_examples.ml` [12] checks an algebraic (Chou-style) encoding of Simson's theorem with the complex Gröbner-basis procedure `COMPLEX_ARITH`; the geometric versions in the same file are commented out as too slow. The main files of our Lean development prove only the forward direction; their `Simson.area2_perpFoot_eq_zero_iff` is a determinant-form statement between local expressions, not the geometric converse proved in Isabelle and Rocq. Version 2 adds the geometric converse in a Layer-15 file (`Simson.Classical.cov_A3_b`, Section 5), after those formalisations; no priority is claimed for it. In Lean, the theorem is not in Mathlib and not in `docs/1000.yaml`; a GitHub code search for "simson" in Lean found one planning file listing "SimsonLine – Define the Simson line" with no proof [8], and otherwise unrelated hits. The original project search missed the Isabelle and Rocq entries; they were identified during revision. As far as we know, ours is the first Lean proof of the forward direction, and nothing more is claimed.

Napoleon. Napoleon's theorem has prior formalisations, and our contribution is not novelty. A Lean 4 repository by Imran Tanrıkolu [18] (Lean `v4.15.0`, 64-line file `napoleantheorem.lean`, no `sorry`) proves in complex coordinates that the three Napoleon points satisfy $\mathrm{IsEquilateral}(z_1, z_2, z_3) : (z_1 - z_2)^2 + (z_2 - z_3)^2 + (z_3 - z_1)^2 = 0$. This is the algebraic criterion for an equilateral triangle; the repository does not state or prove the equality of distances, and treats one rotation only. Its core is the same kind of `linear_combination` certificate as (3). The AFP entry *Napoleon's Theorem* by Ramos, Hulak and de Queiroz [16] (2026) formalises the theorem in complex coordinates, including the exterior-side condition and both orientations of the rotation. The benchmark repository `epfl-lara/icml-26-lean-challenges` [17], file `autformalization/problems/geometry/L2/geo_gen_L2_006/ShadowBench/Source/Main.lean` at commit `7541fcec` (12 June 2026), contains complete Lean proofs of both the inner and the outer Napoleon distance equalities, including side conditions (around lines 269 and 355); this is a second earlier Lean proof, and it already states equality of distances for both triangles. A further, independent Lean proof was published *after* ours: the repository `facebookresearch/atlas-lean` [4], file

MathlibExt/Geometry/Euclidean/NapoleonTheorem.lean, added by pull request #63 at commit 27a38cfdcea9e80a11c49ff8e2ce17650e264552 (5 October 2026, 21:40 UTC; 400 lines, no sorry), proves the outer Napoleon theorem in EuclideanSpace ℝ (Fin 2) (napoleon_theorem_general and napoleon_theorem), with the apexes given by distance equalities and the exterior condition by signed areas, and shows that the common distance is positive. Our Napoleon development was made in early October 2026 and made public at github.com/jpbald93/napoleon on 5 October 2026 (repository created 17:16 UTC), a few hours before the Atlas file became public, and independently of it. As for Monge, this is a statement about public dates only: the Atlas file is migrated from the private repository fairinternal/formal-math, so earlier private authorship cannot be excluded, and we claim no priority over it. Guilhot's Coq development of plane geometry with oriented angles also treats Napoleon's theorem (file point_napoleon.v) [10, 9]. In Mathlib, docs/1000.yaml lists Napoleon's theorem with no declaration. Relative to the Tanrıkolu repository, our version states equality of distances, proves both Napoleon triangles and the orientation lemma; relative to the ShadowBench proof and the AFP entry, it adds nothing of substance beyond its hypothesis analysis.

Lami. docs/1000.yaml lists Lami's theorem with no declaration. GitHub code search for "Lami theorem" and "lami" in Lean (51 and 68 hits) returned only false positives, repository search found one Java and one HTML project, and a search of Mathlib issues and pull requests returned nothing. Mathlib has the law of sines (EuclideanGeometry.law_sin, InnerProductGeometry.sin_angle_mul_norm_eq_sin_angle_mul_norm) but no statement about equilibrium of forces.

Summary. We found no earlier formal proof of Lami's theorem, in Lean or in other systems; given that our searches missed existing Monge, Simson and Napoleon work, this is a report of what we found, not a claim of priority. Monge's three-circle theorem has two earlier independent Lean proofs, one predating ours by four months; our version differs only in its arbitrary-field formulation and hypothesis analysis. The Desargues development is, as far as we know, the first Lean proof of the projective-plane formulation; the theorem is formalised in at least five other systems, including a coordinate proof in HOL Light, and an unmerged Mathlib pull request proves the affine (parallel) form. The Simson development is a Lean proof of the forward direction (version 2 adds the converse) of a theorem already formalised, with its converse, in Isabelle and in Rocq. Napoleon's theorem is formalised in Lean (twice), Isabelle and Rocq already. The searches are not exhaustive, and the Monge, Simson and Napoleon cases show that they missed existing work.

10 Verification

The battery results below were produced on the build machine on 5 October 2026. The two author-check confirmations have since been supplied and their placeholders resolved; no placeholders remain in the paper. The version-2 commits are published at the hashes given in the availability paragraph. The Layer-14 and Layer-15 checks of version 2 are described in Section 10.1.

What has already been checked. Each project report records a full lake build with 0 errors and 0 warnings; on the build machine the cited commits build with 0 errors and 0 warnings in 1,805 jobs (Desargues), 3,013 (Monge), 8,711 (Simson), 8,709 (Napoleon) and 2,428 (Lami). The project reports also record an empty grep of the library sources for sorry, admit, native_decide, axiom, #eval, run_cmd, set_option, macro, elab, syntax, notation and import Lean, and #print axioms output, from a file kept outside the library, showing only propext, Classical.choice and

`Quot.sound` for 17 (Desargues), 3 (Monge), 4 (Simson), 5 (Napoleon) and 15 (Lami) declarations; the gate below covers all cited declarations, including the bridge lemmas. On 5 October 2026 the library sources, `gate.sh`, `tests/tamper.sh` and Lake files of the cited commits were compared by SHA-256 with the copies on the build machine and agree. The five repositories were also rebuilt from fresh clones of the published commits on a second machine with a different architecture and operating system (arm64, macOS): the builds are warning-free, `gate.sh` passes in all five at the same declaration counts, and `tests/tamper.sh` rejects all six defects. Python scripts in each `code/` directory check the statements numerically: for Desargues, 3,000 random perspective configurations over $\mathbb{Q}$ and about 600,000 configurations over $\mathbb{F}_5$ including degenerate ones; for Monge, 2,000 random configurations over $\mathbb{Q}$; for Simson, 2,000 random configurations with an off-circle negative control; for Napoleon, 1,000 random triangles with two negative controls; for Lami, 100,000 random equilibrium triples. None of these scripts is part of the formal proof.

House battery. The same battery as for the earlier AFM papers was run on the five repositories (reports of 5 October 2026, 20:42–21:27 UTC). The build machine's copies are not git checkouts, so the reports record no commit; their Lean sources, `gate.sh`, `tests/tamper.sh` and Lake files are SHA-256-identical to those of the commits cited above, so the results apply to those commits.

- *Gate.* `gate.sh` in each repository rejects the forbidden tokens, builds the library, generates its own `#print axioms` query for every declaration cited in this paper, and fails on any non-standard axiom. Result: PASS, standard axioms only, in all five: 24 declarations (Desargues), 7 (Monge), 12 (Simson), 21 (Napoleon), 17 (Lami).

- *Tamper tests.* `tests/tamper.sh` plants six defects in scratch copies (a `sorry`; a split `axiom`; a fake `#eval` report; a macro; a deleted headline hypothesis; a wrapped extra axiom) and requires each to be rejected. Result: 6/6 rejected in each of the five repositories.

- *Conclusion and hypothesis mutations.* Small edits to conclusions and hypotheses of each cited theorem, with the proof kept, must fail to compile; survivors are classified by hand. Result: mutants rejected: Desargues 57/58 (2 by heartbeat timeout), Monge 16/16, Simson 29/29 (4 by heartbeat timeout), Napoleon 47/47, Lami 44/44. The one survivor, in `Desargues.ex_nonvacuous`, replaces $\neq 0$ by $\neq 1$ in the last conjunct; the value there is $[a'b'c'] = 58$, which is neither 0 nor 1, so the mutant is an equivalent (still true) statement and its survival is expected. The Desargues run also covered a scratch file on the build machine that is not in the repository (`dt1.lean`, a copy of the bracket identity); it contributes one mutant (rejected by timeout), one statement and one ablation to the counts below. Without it the Desargues figures are 56/57 mutants rejected (1 by timeout), 10 hypothesis sets probed, 50 ablations NEEDED and 20/23 statements certified.

- *Contradiction probe.* Automation tries to derive `False` from the hypotheses of each theorem, with a control that must compile. Result: hypothesis sets probed: 11 (Desargues), 2 (Monge), 10 (Simson), 4 (Napoleon), 14 (Lami), the remaining statements having no hypotheses; 0 found contradictory.

- *Hypothesis ablation.* Each Prop hypothesis is replaced by `True` in a renamed copy, which must then fail to compile. This shows that the proof uses the hypothesis, not that the weakened statement is false; for that, see the remarks in Sections 3–7. Result: NEEDED 51 (Desargues), 5 (Monge), 24 (Simson), 7 (Napoleon), 53 (Lami); UNUSED 0 and SKIP 0 in all five; no failed controls. In particular the new hypotheses $r_1 \neq 0$, $r_1 \neq r_2$ of `Monge.ext_eq_`

`lineMap_homothety` and area2$(a, b, c) \neq 0$ of `Simson.concyc_eq_zero_iff_cospherical` are used by their proofs.

- *Unused-variable linter.* Every library file is compiled with `linter.unusedVariables` enabled, with a positive control. Result: 0 warnings, in the library files and in the scratch files of all five repositories.

- *Satisfiability.* For each hypothesis-bearing statement, a kernel-checked `example` exhibits concrete witnesses for all hypotheses. The non-vacuity examples of Sections 3–6 already cover the headline statements of Desargues, Monge, Simson and Napoleon; Lami has none in the library. Result: certified/total: Desargues 21/24, Monge 5/5, Simson 14/16, Napoleon 19/22, Lami 14/17; every new bridge lemma is certified. Unresolved, and not counted as success: `Desargues.desargues_mk`, `Desargues.desargues_converse_mk`, `Desargues.br_eq_zero_of_isCollinear_mk`; `Simson.simson`, `Simson.simson_area`; `Napoleon.apex_equilateral`, `Napoleon.key`, `Napoleon.napoleon_general`; `Lami.lami_sin_ne_zero`, `Lami.lami_nondegenerate`, `Lami.lami_oriented_converse`.

- *Bridge inventory.* Every local `def` is listed with its bridge to a Mathlib notion, if any (Section 8). Result (heuristic: a notion counts as bridged if some lemma mentioning it is found): Monge 2/2 bridged (`Monge.det3`, `Monge.ext`); Simson 4/4 (`Simson.footParam`, `Simson.perpFoot`, `Simson.area2`, `Simson.concyc`); Napoleon 6/6 (`Napoleon.centroid`, `Napoleon.apex`, `Napoleon.signedArea`, `Napoleon.Ga`, `Napoleon.Gb`, `Napoleon.Gc`); Lami has no local definitions. For Desargues the scanner reports 11 notions, 8 bridged (the bracket and the example points O, A, B, C, P, Q, R) and 3 not (`exA'`, `exB'`, `exC'`). That report is an artefact of the scanner: it searches for the notion's name with a word-boundary pattern `\b`*name*`\b`, which cannot match a name ending in a prime, so `exA'`, `exB'`, `exC'` are never found even though `Desargues.exA'_isCollinear`, `Desargues.exB'_isCollinear` and `Desargues.exC'_isCollinear` mention them. By inspection, all 11 Desargues notions have bridge lemmas. The two Monge and Simson bridges carry the hypotheses stated in Sections 4 and 5 ($r_1 \neq 0$; area2$(a, b, c) \neq 0$), and Simson's `Simson.area2` is bridged in one direction only at the cited commits (the other direction is in the Layer-15 file of version 2).

Automated checks cannot replace comparing the formal statements with their informal counterparts. The author has checked each entry of Table 1 by hand.

10.1 Layers 14 and 15

Version 2 adds two further checks, in two new files of each library, `L14.lean` and `L15Classical.lean` (for example `Desargues/L14.lean`, `Desargues/L15Classical.lean`). The library files of the cited commits are unchanged (SHA-256). With the new files the gates pass with 34 declarations (Desargues), 15 (Monge), 21 (Simson), 45 (Napoleon) and 23 (Lami), standard axioms only, and `tests/tamper.sh` rejects all six defects in each repository (arm64, macOS).

Layer 14: Mathlib-only statements and non-vacuity. Each of the six numbered theorems has a Mathlib-only formulation in `L14.lean`, proved from the library theorem through bridges; for Napoleon it covers only non-degenerate configurations (see below). A mechanical check of the elaborated types, also with all implicit and instance arguments shown, found no constant of the library's namespace. Each restatement has a witness theorem that proves every hypothesis on an explicit configuration and that the configuration is non-degenerate:

- Theorem 1: `Desargues.l14_desargues_det` (bracket as `Matrix.det`, Mathlib's cross product); witness `Desargues.l14w_desargues_det`: both triangles non-degenerate, the three Desargues points computed and pairwise non-proportional.
- Theorem 2: `Desargues.l14_desargues`, quantified over points of $\mathbb{P}(K^3)$ rather than representatives, perspective as `Projectivization.IsCollinear`; the hypotheses "representative $\neq 0$" disappear. Witness `Desargues.l14w_desargues` over $\mathbb{Q}$: neither triangle collinear, O distinct from the six vertices, $A \neq A'$, $B \neq B'$, $C \neq C'$, the three Desargues points pairwise distinct.
- Theorem 3: `Monge.l14_monge`, each centre of similitude given by its defining property $r_1(O_2 - S) = r_2(O_1 - S)$, conclusion `Collinear`, with exactly the paper's hypotheses (and a homothety form `Monge.l14_monge_homothety`, which needs nonzero radii for the ratio); witness `Monge.l14w_monge`: the example of Section 4, centres not collinear, the three centres of similitude pairwise distinct.
- Theorem 4: `Simson.l14_simson`, feet as `EuclideanGeometry.orthogonalProjection`; witness `Simson.l14w_simson`: $a = 5$, $b = 3 + 4i$, $c = -4 + 3i$, $p = -5i$, triangle not collinear, p not a vertex, the three feet computed and pairwise distinct.
- Theorem 5: `Napoleon.l14_napoleon_outer` and `Napoleon.l14_napoleon_inner`, apexes by equal sides and strict opposite (same) side, centres by `Finset.centroid`, no orientation hypothesis; witness `Napoleon.l14w_napoleon`: $A = 0$, $B = 2$, $C = 2i$, the three centres computed and pairwise distinct.
- Theorem 6: `Lami.l14_lami`, the library statement being already Mathlib-only (and `Lami.l14_lami_oriented`); witness `Lami.l14w_lami`: forces 3, $4i$, $-3 - 4i$, pairwise distinct and nonzero, at a right angle, common ratio 5.

Layer 14 gives Mathlib-only formulations associated with all six numbered theorems. For Napoleon, the strict-side formulation covers non-degenerate geometric configurations in either orientation; it is not a restatement of the degenerate cases allowed by Theorem 5, which remain covered by the original rotation-based statements. The secondary forms (`Napoleon.l14_napoleon_inner`, `Lami.l14_lami_oriented`, `Monge.l14_monge_homothety`, `Simson.l14_simson_feet`) have no separate witness, and the converses of Table 1 are not restated in Layer 14 (the Desargues converse is covered by Layer 15). The witnesses give concrete configurations for the Mathlib-only forms of `Desargues.desargues_mk`, `Simson.simson` and `Lami.lami_nondegenerate`, which the satisfiability check above left unresolved; the battery figures above refer to the cited commits.

Layer 15: comparison with the classical statements. Another AI agent collected textbook statements of the five theorems from the literature, with citations (Coxeter; Coxeter–Greitzer; Johnson; encyclopedic sources): Desargues and its converse (A1), Monge with its companion form (A2), Wallace–Simson with its converse (A3), the outer and inner Napoleon triangles (A4) and Lami (A5). An AI model given only those statements, which saw no code and not this paper, wrote them in Lean with Mathlib notions, with its own definitions only where Mathlib has none (centres of similitude and equilateral apexes by their defining properties): 9 theorems and 5 construction obligations, all with `sorry`. Its transcript was audited: a transcript audit in the verification repository found no file read beyond its own brief and outputs and no web access. It marked four ambiguities and one extra hypothesis (positive radii, since Mathlib's spheres allow radius ≤ 0), and stated A3 in `EuclideanSpace` $\mathbb{R}$ `(Fin 2)`. A comparer then proved each of the 14 statements, copied

verbatim, from the libraries at the cited commits without changing them, or classified it: 9 covered by the library theorems (A1(a), A1(b), A2(a), A2(b), A3(a), A4(a), A4(b), A5 and its obligation); 4 outside the paper's statements but proved (the obligations of A1 and the two of A2, and the Simson converse A3(b)); none narrower than the library; 1 writer statement false as written; none open. The writer's hypotheses $A \neq A'$, $B \neq B'$, $C \neq C'$ in A1(a) and distinct centres in A2(a) were not needed; the latter agrees with Table 2. New proofs, not in the main files:

- `Simson.Classical.cov_A3_b`: the geometric converse of the Wallace–Simson theorem, through the new bridge `Simson.Classical.area2_eq_zero_of_collinear` (Section 5).
- `Napoleon.Classical.cov_A4_a`: apexes by Mathlib's half-planes, and the outer Napoleon triangle is affinely independent, by an identity for its doubled signed area.
- `Monge.Classical.cov_A2_b`: the companion form as a declaration; the paper had only the instantiation $r \mapsto -r$ in prose.
- `Desargues.Classical.cov_A1_obligation`: two distinct lines of $\mathbb{P}^2(K)$ meet in exactly one point.
- `Monge.Classical.cov_A2_obligation_ext`, `Monge.Classical.cov_A2_obligation_int`: existence and uniqueness for two circles with positive radii and distinct centres, with unequal radii additionally required for the external centre.
- `Napoleon.Classical.cov_A4_obligation_corrected`: on each side of a segment with $U \neq V$ there is exactly one equilateral apex.

Narrower cases: none. Writer mis-statement (one): `classical_A4_obligation` omits $U \neq V$; for $U = V$ the only apex is U itself, on the "line", so no apex is strictly on either side. The negation is proved (`Napoleon.Classical.A4_obligation_false`) and the corrected statement above. This is an error of the writer, not of the paper. The comparer also noted, as Section 6 says, that the library's ω_o apexes are outward only for counter-clockwise triangles, so the clockwise case of A4 uses `Napoleon.napoleon_inner`. All these proofs depend only on the three standard axioms. They are collected in `L15Classical.lean` of each repository (namespace `Desargues.Classical` and so on), whose gate lists every one of them; the classical definitions and statements are reproduced there verbatim.

11 Limitations and further work

- *Monge stops at a determinant.* The conclusion of `Monge.monge` is $\det_3 = 0$ for explicit formulas. The bridges `Monge.det3_eq_zero_iff_collinear` and `Monge.ext_eq_lineMap_homothety` identify $\det_3 = 0$ with `Collinear` and, for $r_1 \neq 0$, $r_1 \neq r_2$, the point `Monge.ext` with the centre of a homothety sending O_1 to O_2, but the main files give no composed statement about circles: there are no circles in the statement, and that the homothety maps one circle onto the other is not proved there. Version 2 adds `Monge.l14_monge`, with conclusion `Collinear`, and proves in the Layer-15 file, for circles with distinct centres and positive unequal radii given as Mathlib spheres, that the homothety maps one circle onto the other at a unique external centre and, for three circles with pairwise distinct radii, that the three external centres are collinear (`Monge.Classical.cov_A2_obligation_ext`, `Monge.Classical.cov_A2_a`). All equal-radius cases are excluded (one or three centres of similitude at infinity, or no unique centre for identical circles; Section 4), and the tangent-line formulation, which needs the external tangents to exist, is not proved.

- *Converses.* In the main files the Desargues converse concludes collinearity in the dual plane, and turning this into the existence of a point on three lines requires a second lemma; the composition is proved in the Layer-15 file (`Desargues.Classical.cov_A1_b`). The main files prove only the forward direction of the Simson theorem, and `Simson.area2_perpFoot_eq_zero_iff` is only a determinant-level statement; the geometric converse (proved in the Isabelle and Rocq developments) was not proved in the first version and is proved in version 2, in the Layer-15 file (`Simson.Classical.cov_A3_b`). The Lami converse is oriented only, as it must be.

- *Fields.* In the main files Simson and Napoleon are stated in $\mathbb{C}$ only, not in a general Euclidean plane (`EuclideanSpace` $\mathbb{R}$ `(Fin 2)` or an oriented two-dimensional affine space); the Layer-15 file proves the classical Simson statements (both directions) and both Napoleon statements in `EuclideanSpace` $\mathbb{R}$ `(Fin 2)` (Section 10.1), and Lami's oriented form needs a two-dimensional oriented space. The Simson certificate itself is a polynomial identity; a version over an arbitrary ordered field, or over an arbitrary field with non-isotropic side vectors, has not been attempted.

- *Orientation in Napoleon.* In the main files "outward" is expressed by the sign of the area of the erected triangles, not by half-planes; the Layer-14 and Layer-15 statements use Mathlib's half-planes (`AffineSubspace.SOppSide`; Section 10.1).

- *Desargues in higher dimension.* Only the plane is treated; no statement in projective space of dimension at least three, and no axiomatic version.

- *Lami and the law of sines.* The force-triangle derivation is not formalised.

- *Novelty.* Monge's three-circle theorem already has two complete Lean proofs, one of them earlier than ours; Napoleon's theorem is formalised elsewhere, including in Lean; the Wallace–Simson theorem and its converse are formalised in Isabelle and Rocq; Desargues's theorem is formalised in several other systems (including by coordinates in HOL Light), and its affine parallel form is the subject of an open Mathlib pull request.

Acknowledgements and use of AI

A large-language-model AI assistant was used to write the Lean proofs and the Python checking scripts, to run the prior-art searches, and to draft this paper. The author directed the work and reviewed the paper. The author checked each formal statement in Table 1 against its informal counterpart by hand. The Layer-14 and Layer-15 files of version 2, the collection of the classical statements, their Lean formulation and the comparisons were carried out by AI agents and are not part of the author's hand check. The resulting formal proofs and counterexample are checked by the Lean kernel; fidelity to the textbook sources and the comparison classifications require separate semantic review. Every proof is checked by the Lean kernel, and the author takes responsibility for all content.

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Theorem	Hypothesis	Role	Without it
Desargues	$o \neq 0$	turns perspective into $[AA'BB'CC'] = 0$	statement false
	$[o\,a\,a'] = [o\,b\,b'] = [o\,c\,c'] = 0$ (forward)	perspective from O; with $o \neq 0$ gives $[(a\times a')(b\times b')(c\times c')] = 0$, the third factor of (1)	no theorem (the hypothesis)
	six vertex inequalities, three side inequalities, nonzero representatives	translation: Mathlib's `cross` computes lines and meets	conclusion about the wrong points
	$[PQR] = 0$, $P, Q, R \neq 0$, joins $\neq 0$ (converse)	hypothesis of the converse; translation	—
	$[abc], [a'b'c'] \neq 0$ (converse)	cancels a factor of (1)	no information
Monge	$r_i \neq r_j$	clears denominators	false in Lean ($x/0 = 0$); point at infinity
	(centres)	not needed	—
	$r_1 \neq 0$, $r_1 \neq r_2$ (bridge `Monge.ext_eq_lineMap_homothety`)	clears r_1 and $r_1 - r_2$	for $r_1 = 0$, $S_{12} = O_1$ is not a centre of similitude
Simson	a, b, c distinct	clears D_i; feet well defined	feet undefined (Lean: $x/0 = 0$)
	cospherical	supplies concyc $= 0$ in (2)	false (negative control)
	$S \neq 0$ (determinant equivalence)	cancels S^2	no information
	$S \neq 0$ (bridge `Simson.concyc_eq_zero_iff_cospherical`)	circumcentre formula has denominator $2S$	false: four distinct collinear points have concyc $= 0$
Napoleon	$\omega^2 - \omega + 1 = 0$	multiplier in (3)	still implied by the two retained norm hypotheses (not independently removable); false for ω violating all three
	$\lvert\omega\rvert = \lvert 1 - \omega\rvert = 1$	translation to distances	redundant given $\omega^2 - \omega + 1 = 0$ (each set implies the other: both say $\omega = e^{\pm i\pi/3}$)
	(non-degenerate ABC)	not needed	—
Lami	$F_1 + F_2 + F_3 = 0$	substitution in (4)	—
	$F_1, F_2, F_3 \neq 0$	ratio forms (`Lami.lami`, `Lami.lami_oriented`, `Lami.lami_nondegenerate`)	ratios meaningless
	$F_3 \neq 0$ only	division-free form `Lami.lami_cross`: cancels $\lVert F_3\rVert$	product form `Lami.lami_mul` still holds
	$F_1 \nparallel F_2$	sines nonzero	ratios hold only via $x/0 = 0$
	(converse, area form)	hypotheses of	no information
	ω's equal, $\omega(F_1, F_2) \neq 0$	`Lami.equilibrium_of_areaForm`	
	(converse, ratio form)	hypotheses of `Lami.lami`	false with unoriented